

HOMEWORK 2

6500-324: DATA MANAGEMENT FOR INFORMATION SYSTEMS (Fall 2022)

©2022 by ecourse.org

Part 1. Multiple Choice or True/False

1. (1 point) What is the best entity-relationship diagram to model students, dorms, rooms for student housing?
A) Dorm -- (1:m) -- Room --(1:m) -- Student
B) Dorm -- (1:m) -- Room --(m:m) -- Student
C) Student -- (m:1) -- Dorm --(1:m) -- Room
D) Drom -- (1:m) -- Student--(m:1) -- Room
2. (1 point) (check all that apply) Identify Objects for Data Modeling: Which of the following are objects for the problem domain of student registration?
A) tuition, financial services, fees, credit hours, registratrar
B) students, courses, instructors, sections, classrooms
C) advisers, departments, buildings, registrations
D) registration time, class cap, class closed, payment due date
E) textbooks, parents, university, registration system, print receipt
3. (1 point) (check all that apply) Identify Objects for Data Modeling: Which of the following are objects for the problem domain of banking systems?
A) ATMs, employees, branches, customers
B) account balance, credit, debit, fraud
C) customer service, help desk, customer satisfaction, validate accounts
D) withdraws, transfers, deposits
E) products, accounts, orders
4. (1 point) Identify Object Attributes: Which of the following are legitimate attributes of Borrower in a college library system?
A) book title
B) borrower type
C) overdue
D) checkouts
E) returns
5. (1 point) (check all that apply) Identify Object Attributes: Which of the following are legitimate attributes of shipments in an online order system?
A) ship date
B) check shipping status
C) delivery address
D) warehouse
E) delivery date
6. (1 point) (check all that apply) Identify Object Relationships: Check all that justify the connection between the two mentioned objects for a student registration system?

- A) students takes classes
B) instructors teach students
C) students goes to classroom
D) departments offer courses
E) instructors belong to departments
7. (1 point) (check all that apply) Identify Object Relationships: Check all that justify a direct relationship between the two mentioned objects in each sentence for a point of sale system (POS)?
A) cashiers checkout items
B) managers balance inventories
C) customer check price
D) cashiers performs transactions
E) employee handle returns
8. (1 point) (check all that apply) Identify Object Relationships: Check all that justify a direct relationship between the two mentioned objects for airline reservation systems?
A) customers make reservations
B) employees print boarding passes
C) customers make payments
D) passengers are assigned to airplane seats
E) flights have destinations
9. (1 point) Identify Object Relationships: A flight has a starting airport, a destination airport, and zero or more pass-over airports. How do you model the relationships between Flight and Airports?
A) create three different relationships respectively for starting, destination, and passover
B) create a gerund between Flight and Airport and use an attribute to indicate starting, destination, and passover
C) create one direct m:m relationship between Flights and Airports
D) create an recursive relationship on Airports
10. (1 point) Which is not a relevant feature of CASE tools?
A) The ability to help draw data models using entity-relationship notations
B) The ability to generate code
C) An information repository
D) Access to a DB via the Internet